KSoR — Knowledge System of Record

Bonus / Standalone References (outside the 70-chapter core track) — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: docs/<slug>/SUMMARY.md + README.md + quiz.md (self-authored fixture, not a live book <Quiz> component).

EN: KSoR is an open-source SDK (panaversity/ksor) that turns institutional knowledge (policies, rules, thresholds) into a governed, provenance-tracked source served identically to humans (website) and AI agents (MCP) — 'Same Truth'.
Roman Urdu: KSoR ek open-source SDK hai jo institutional knowledge (policies, rules, thresholds) ko governed, provenance-tracked source banata hai — insaan (website) aur AI agents (MCP) dono ko ek hi 'Same Truth' serve karta hai.

00 — Kyun Aur Definition

- Traditional System of Record 'abhi kya sach hai' batata hai (ledger jeetta hai, spreadsheet nahi); KSoR 'hum kya jaante hain, kaise operate karna hai' batata hai.
- Knowledge base sirf **store** karta hai; KSoR **authority** establish karta hai — ownership, provenance, versioning, review, conflict-handling, abstention sab architecture ka hissa hain, optional nahi.

01 — 7 Principles + Kya Bana Sakte Hain

- 'One Source, Two Surfaces': Human Surface (website) aur Agent Surface (MCP) — dono ek hi governed source ke projections.
- 7 principles: One Authoritative Source, Humans+Agents First-Class, Provenance Matters, Citation Before Confidence, Abstention Is a Feature, Governance Before Retrieval, Vendor Neutrality.
- 3 types: Organizational KSoR, Domain KSoR, Product/Method KSoR — Vertical KSoR sirf ek application hai.

02 — Architecture Aur Tooling

- CLI: **init, dev, build, serve** — abhi sirf design hain (package **0.0.0**), implement nahi hue, exit code 2 dete hain.
- Project structure: knowledge/ (plain Markdown corpus), instance.md, site/, .agents/skills/.
- Knowledge as Code pipeline: authored→reviewed→version-controlled→validated→tested→built→published→consumed. **build.lock.json** se Build Provenance chain: AI Answer → Passage → Document → Build → Git Commit → Source.
- Agent Surface = MCP (model-agnostic). KSoR vs RAG: RAG 'context mein kaise dalein', KSoR 'kaunsi knowledge itni authoritative hai ke operate kiya ja sake' — RAG KSoR ka chhota hissa hai.

03 — Governance Aur AI-Native Role

- Governance flow: Source → Draft → Review → Approved → Authoritative → (Human+Agent Surface) → Superseded.
- 3 knowledge-boundary types: in-scope (jawab+source), requires-reasoning (governed rule + operational facts jorna), outside-the-KSoR (guess nahi, mana kar dega).
- Agent-First Development: repo khud knowledge ka artifact bhi hai aur agents ke kaam karne ki working context bhi. Human-readable by default, vendor-free by design.

04 — Applications, Design Goals, Status

- Agent Factory KSoR (shared method) + Vertical KSoR (domain truth) mil kar domain-specific agents kaam karte hain — same ksor SDK se.
- 11 design goals: Authoritative, Governed, Traceable, Inspectable, Portable, Agent-readable, Human-readable, Versioned, Reproducible, Composable, Extensible.
- Status: active development, Apache 2.0, npm package @panaversity/ksor — MCP agent surface abhi sirf design hai.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
System of Record (SoR)	Authoritative source for current operational state.	Abhi ka operational sach batane wala authoritative source.
KSoR	Governed source of institutional knowledge served to humans and agents.	Institutional knowledge ka governed source, insaan+agent dono ke liye.
Same Truth	Human and agent surfaces are projections of one governed source.	Human aur agent surface ek hi governed source ke projections.
Provenance	Traceable record of where knowledge came from and its version.	Knowledge kahan se aayi aur kaunsi version — traceable record.
Abstention	Saying 'not covered' instead of guessing when policy is silent.	Policy na ho to guess nahi, 'maloom nahi' kehna.
Citation Before Confidence	An answer must be traceable, not just stated confidently.	Jawab trace hona chahiye, sirf confident lehja proof nahi.
Governance Before Retrieval	Structure/approval comes before search/serving.	Retrieval se pehle structure/approval zaroori.
Vendor Neutrality	Not locked to one AI provider or platform.	Kisi ek AI vendor/platform se bandha na hona.
Organizational KSoR	Company-wide knowledge (e.g., Agent Factory KSoR).	Poori organization ki knowledge (jaise Agent Factory KSoR).
Domain KSoR	Field-specific knowledge (Accounting, Healthcare, Legal).	Khaas field ki knowledge (Accounting, Healthcare, Legal).
Vertical KSoR	One application of Domain KSoR for a specific vertical.	Domain KSoR ki ek application, khaas vertical ke liye.
build.lock.json	File recording included docs, hashes, source commit, version.	Included docs, hashes, source commit, version record karne wali file.
Build Provenance chain	Answer → Passage → Document → Build → Commit → Source.	Jawab se source tak traceable chain.
Knowledge boundary	In-scope / requires-reasoning / outside-the-KSoR classification.	In-scope / requires-reasoning / outside-the-KSoR classification.
MCP (Agent Surface)	Model-agnostic interoperability boundary for agents.	Agents ke liye model-agnostic interoperability boundary.

Viva Prep — Scenario Questions (from quiz.md, 12 total)

#	Scenario Gist	Answer	One-line Reasoning
1	Spreadsheet $1.2M vs accounting ledger $1.15M — kaun jeetta hai?	**Ledger jeetta hai — wahi authoritative Traditional SoR hai.**	Governed, reviewed, audit-trailed source hamesha convenient-but-ungoverned source se jeetta hai.
3	RAG + vector DB ko team 'yeh humara KSoR hai' kehti hai — kya missing hai?	**Authority — ownership, provenance, versioning, review, conflict-resolution, abstention.**	Storage/retrieval kaafi nahi; KSoR authority establish karta hai, RAG sirf implementation detail ho sakta hai.
5	Agent ko policy clear nahi — cost capitalize karna hai ya nahi?	**Bata de ke policy is situation ko cover nahi karti — khud se policy invent na kare.**	Abstention Is a Feature: guess (conservative ya aggressive dono) authority ko replace kar deta hai.
8	`npx @panaversity/ksor init` chalane par kya hoga?	**Status notice print hoga, exit code 2 — commands abhi sirf design hain (package 0.0.0).**	Doc se pehle verify karo SDK implement hui ya sirf designed — 'test before claiming done' reminder.
10	Team KSoR ko 'bas RAG system hai' keh kar dismiss karti hai — sabse bari galti?	**RAG = 'context mein kya dalein'; KSoR = 'kaunsi knowledge authoritative hai ke operate kiya jaye' — retrieval chhota hissa hai.**	KSoR mein Governance, Authority, Provenance, Versioning, Review, dono surfaces shamil hain.
12	Agent se poocha 'agle saal ki approval policy kya hogi?' — kaunsi category?	**Outside the KSoR — approved nahi hai (future), seedha mana kar dena, guess nahi.**	3 boundary types mein future/unapproved cheez seedha category 3 mein aati hai.

Where This Leads / Sources

Agle qadam: KSoR ke 7 principles aur Vertical SoR design method ([[ecosystem-designing-the-vertical-sor]]) ka moazna karo — dono independent hain magar overlapping concerns (governance, provenance, abstention) share karte hain. CCAR-F FDE Track B practicum isi pattern ka real scaffold banata hai.

docs/ksor/SUMMARY.md, README.md, 00-04 numbered files, quiz.md, 05-test-your-understanding.md (panaversity/ksor GitHub README + KSoR-Complete-Guide PDF se grounded).